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Modeling of Polyatomic Gases in the Kinetic Fokker-Planck Method by Extension of the Master Equation Approach

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Abstract. An extension of the Fokker-Planck method for polyatomic gas using the master equation approach is presented. The model is compatible with different Fokker-Planck formulations. An example using the cubic formulation is provided. Code-to-code comparison between the new model and the DSMC implementation in the *PicLas* code of the university of Stuttgart is presented using a heat bath temperature relaxation as well as two dimensional, non-reacting, hypersonic flow over a flat-faced cylinder. The heat bath test case shows very good agreement between the DSMC and Fokker-Planck implementation. The hypersonic test case compares very well qualitatively but as expected shows deviations in areas of strong thermal non-equilibrium.

INTRODUCTION

Flows encountered in space applications, like atmospheric reentry or thruster plume expansion and interactions, can be described by the well-known Boltzmann equation. A common approach to numerically solve this equation is the Direct Simulation Monte-Carlo (DSMC) method pioneered by Bird [1]. The method is very efficient for high Knudsen numbers but becomes increasingly computationally intensive when approaching the continuum limit. In this regime the Boltzmann equation can be numerically solved using the recently proposed kinetic Fokker-Planck (FP) method [2] which, like DSMC, relies on simulated particles to transport mass, momentum and energy through the flow domain, but does not resolve individual collisions. Due to the similarity in their formulation a hybrid method based on DSMC and FP can be derived which allows a computationally efficient simulation of flows with a broad range of Knudsen numbers [3].

Modeling flows of gas with internal degrees of freedom, such as for example CO₂, is essential for space engineering applications like studying entry missions into the atmosphere of Mars [4]. The FP model has been extended to describe diatomic gases for single species [5, 6] as well as for mixtures [7, 8]. An extension to treat polyatomic gas has been proposed by Mathiaud et. al. [9]. However, this approach uses a single scalar value to model the internal energy of the gas and thus does not capture the distribution of internal energy between the individual degrees of freedom. An approach to resolve these internal degrees of freedom is described in this paper.

METHODS

Cubic Fokker-Planck

We base our further discussion on the cubic formulation of the Fokker-Planck method, which is covered in detail in the literature [6, 10, 11] and thus will be only briefly summarized here.

The well-known Boltzmann equation describes the evolution of a scalar distribution function f due to intermolecular collisions in rarefied flows:

$$\frac{Df}{Dt} = S_{\text{Boltz}}, \quad (1)$$

where t is time and S_{Boltz} is the Boltzmann collision integral. A recently introduced approach [2] to make the integro-differential Eq. (1) more manageable is to approximate S_{Boltz} by a Fokker-Planck collision operator S_{FP} :

$$S_{\text{Boltz}} \approx S_{\text{FP}} = -\frac{\partial}{\partial v_i}(A_i f) + \frac{\partial^2}{\partial v_j \partial v_j} \left(\frac{D^2}{2} f \right), \quad (2)$$

where the drift coefficient A_i and the diffusion coefficient D of Eq. (2) are model parameters chosen in such a way that moments calculated using the Boltzmann collision operator are reproduced in the continuum limit. Here v represents

the velocity coordinate. Note that Einstein's summation convention is used throughout this paper. Parenthesized indices imply as suppression of the convention. According to Ito's lemma Eqn. (2) can be written as stochastic differential equation

$$\frac{d\mathbf{X}}{dt} = \mathbf{V} \quad (3)$$

$$\frac{d\mathbf{V}}{dt} = \mathbf{A} + D \frac{d\mathbf{W}}{dt} + \mathbf{F} \quad (4)$$

with $\mathbf{X}$ denoting particle position, $\mathbf{V}$ the particle velocity and $d\mathbf{W}$ the Wiener process with zero expectation. Thus, the resulting Fokker-Planck equation can be solved through stochastic motion which in turn can be modelled using a particle method.

For a single species gas using the cubic model the drift A_i and the diffusion coefficient D are chosen as

$$A_i = -\frac{1}{\tau} c_i + \xi_{ij} c_j + \gamma_i \left(c_j c_j - \frac{3k_B T}{m} \right) + \Lambda \left(c_i c_j c_j - \frac{2q_i}{\rho} \right), \quad (5)$$

$$D = \sqrt{\frac{2k_B T}{\tau m}}. \quad (6)$$

where τ is a relaxation time, γ and ξ are the coefficient matrices, $\mathbf{c}$ the velocity vector, k_B is the Boltzmann constant, T the temperature, m is mass, $\mathbf{q}$ the heat flux vector and ρ the (mass) density. The parameter Λ is chosen to assure stability of the model. Eqn. (4) can be integrated leading to

$$\mathbf{V}^{n+1} = \frac{1}{\alpha} \left[\mathbf{C}_i^n \exp\left(-\frac{\Delta t}{\tau}\right) + \left(1 - \exp\left(-\frac{\Delta t}{\tau}\right)\right) \tau \mathbf{N}_i^n + \psi_i \sqrt{\frac{D^2}{2} \tau \left(1 - \exp\left(-2\frac{\Delta t}{\tau}\right)\right)} \right] + V_i^n - C_i^n \quad (7)$$

where $\mathbf{C}$ denotes the thermal particle velocity and ψ_i are independent normally distributed random numbers with a mean of 0 and a standard deviation of 1. The nonlinear term $\mathbf{N}$ can be expressed as

$$N_i^n = \xi_{ij} C_j^n + \gamma_i \left(C_j^n C_j^n - \frac{3k_B T}{m} \right) + \Lambda \left(C_i^n C_j^n C_j^n - \frac{2q_i}{\rho} \right). \quad (8)$$

The factor α on Eq. (7) is used to scale particle velocities to ensure energy conservation for arbitrary time steps. To be compatible with DMSC a simple integration scheme for particle positions is chosen as

$$\mathbf{X}^{n+1} = \mathbf{X}^n + \mathbf{V}^n \Delta t. \quad (9)$$

Master Equation Approach

The basic premise of master equation approach [6] is the fact that the Boltzmann collision operator can be split into an elastic ($S_{\text{Boltz}}^{\text{el}}$) and an inelastic part $S_{\text{Boltz}}^{\text{inel}}$:

$$S_{\text{Boltz}} = S_{\text{Boltz}}^{\text{el}} + S_{\text{Boltz}}^{\text{inel}}. \quad (10)$$

The goal is to approximate both parts in such a way as to stay compatible with the Fokker-Planck formulation. The elastic operator is formulated as before, but only a fraction of the particles undergo the elastic collisions. The inelastic part may again be split into two parts:

$$S_{\text{Boltz}}^{\text{inel}} = S_v + S_e \quad (11)$$

where S_v describes the change of the number of particle in a phase space element due to a change of particle velocities and S_e refers to the change in the number of particles in a phase space element due to a change of internal energies as a result of inelastic collisions. The first part can be modeled via the Fokker-Planck approach described above, where the diffusion coefficient is modified to account for energy transfer between translational and internal degrees of freedom. The second part is modeled using the master equation approach as

$$S_e = \sum_j (R_{(jn)} f_j - R_{(nj)} f_n). \quad (12)$$

Here R_{qk} are the rate coefficients modeling the transition rates between the internal energy states q and k . When combined with the original FP equation the new FP-Master Equation can be written as

$$\frac{Df_n}{Dt} = \tilde{S}_{FP}(f_n) + \sum_j (R_{(jn)}f_j - R_{(nj)}f_n). \quad (13)$$

where $\tilde{S}_{FP}(f_n)$ is the FP operator with the modified diffusion coefficient. Assuming that the evolution of translational velocities and internal energies can be modelled as independent stochastic processes:

$$f_n(\mathbf{r}, \mathbf{v}, t) = f(\mathbf{r}, \mathbf{v}, t) \cdot g_n(\mathbf{r}, t), \quad (14)$$

Eqn. (13) can be separated into a set of equations

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial x_i} v_i = \tilde{S}_{FP} \quad (15)$$

$$\frac{dg_n}{dt} = \frac{\partial g_n}{\partial t} + \frac{\partial g_n}{\partial x_i} v_i = \sum_j (R_{(jn)}g_j - R_{(nj)}g_n), \quad (16)$$

which are to be solved independently in each time step [6].

Models for Rate Coefficients

The transfer of energy between the internal degrees of freedom is implicitly modeled in the rate coefficients. The coefficients can be divided into two groups: discrete and continuous. Hepp *et al.* [6] propose two types of coefficients based on the Landau-Teller and the Larsen-Borgnakke relaxation.

Landau-Teller Relaxation

The Landau-Teller relaxation [12] describes a simple energy relaxation process as seen in Eq. (17)

$$\frac{de}{dt} = \frac{1}{\tau} (e^{\text{eq}} - e), \quad (17)$$

where e is the current internal energy, e^{eq} the equilibrium energy, and τ is the characteristic relaxation time of the process. This approach was applied to model the relaxation of continuous energy distribution (continuous Landau-Teller - CLT) as well as discrete (discrete Landau-Teller - DLT).

DLT For the discrete Landau-Teller model the internal structure of the molecule is modeled as a harmonic oscillator. Assuming a Boltzmann distribution of the integral energy states, the partition function can be defined as

$$Q = \sum_i \exp(-\beta e_i) = \frac{1}{1 - \exp(-\beta \epsilon)}. \quad (18)$$

Here, $e_i = \epsilon i$ is the energy of the i th internal energy level, $\epsilon = k_B \theta$ is the spacing between two adjacent energy levels expressed through the characteristic temperature θ , and $\beta = 1/(k_B T)$.

The rate constant can be expressed as

$$R_{ij} = R_j = \frac{\exp(-\beta e_j)}{\tau_{\text{int}} Q}, \quad (19)$$

where $\tau_{\text{int}} = Z_{\text{int}}/v_{\text{int}}$ is the internal relaxation time that can be calculated using the equilibrium mean collision frequency v_{int} and collision number Z_{int} .

CLT For the CLT rate constant the internal energy of the molecule is modeled as a continuous scalar e . Here the rate constant can be written as

$$R(e, e') = R(e') = \frac{\beta \exp(-\beta e')}{\tau_{\text{int}}}, \quad (20)$$

with e' denoting the energy state after transition.

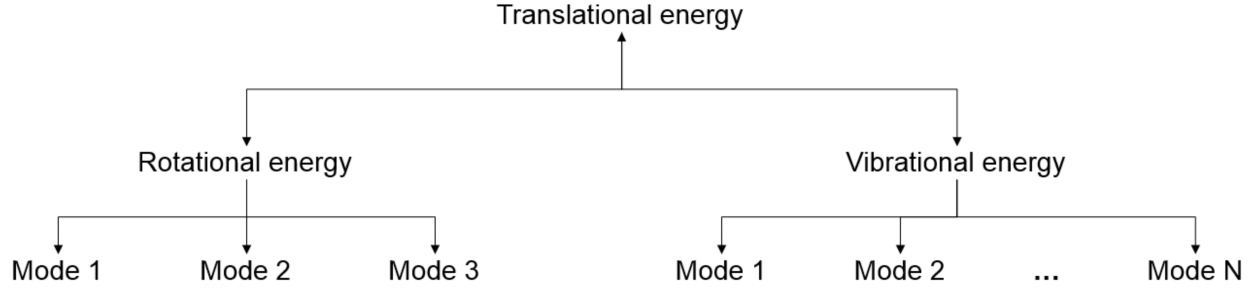


FIGURE 1. Possible energy transfer paths.

Larsen-Borgnakke

Originally derived for continuous energy distribution [13] and later extended for discrete [14], the Larsen-Borgnakke method can be applied to derive rate coefficients for the master equation approach. The model is specifically based on the work of Bergemann [15] and Zhang [16]. In the discrete Larsen-Borgnakke (DLB) model the rate constant R depends on both start (i) and end (j) energy state and must be divided into a forward R_{ij} and backward transition R_{ji} . For the former, the rate constant is defined as

$$R_{ij} = v_{VHS}(T, n) \cdot P_{inel} \cdot \psi\left(i, j, \frac{T}{\theta}\right). \quad (21)$$

In Eq. (21), P_{inel} is the probability for an inelastic collision. The reverse transition can be defined as

$$R_{ji} = R_{ij} \cdot \exp(-\beta \cdot (e_i - e_j)), \quad (22)$$

where ψ is a function that depends on i, j and the fraction T/θ :

$$\psi\left(i, j, \frac{T}{\theta}\right) \equiv \frac{1}{\Gamma\left(\frac{1}{2} - \omega\right)} \int_{\frac{\theta}{T}(i-j)}^{\infty} \exp(-\chi) \chi^{\frac{3}{2}-\omega} \left(1 - \frac{j}{\frac{T}{\theta}\chi + i}\right)^{\frac{3}{2}-\omega} \frac{1}{Q_{nom}} d\chi. \quad (23)$$

Here Q_{nom} is a scaling factor chosen in such a way that $\sum R_{ij} = P_{inel}$. To speed up the simulation Eq. (23) is pre-calculated and stored in a look-up-table.

Extension of the Master Equation Approach to Polyatomic Gases

For the extension of the master equation approach to polyatomic molecules the model must be able to handle an arbitrary number of degrees of freedom. To achieve this goal two assumptions are made:

1. No direct energy exchange between the individual modes of the rotational and vibrational degrees of freedom.
2. No direct energy exchange between the rotational and vibrational energy.

An indirect energy transfer between all modes is still possible through the kinetic energy of the particles. The possible transfer paths are shown in Figure 1. In this first approximation the calculation of coefficients of the Fokker-Planck equation remain unchanged compared to the diatomic case. The basic idea proposed by Hepp [6] remains and is displayed in Algorithm 1: the code iterates over each cell for which the temperature T and the collision frequency ν is calculated. After that the contribution for both the rotational and vibrational relaxation to the new diffusion factor K is calculated as K_{rot} and K_{vib} , respectively. After the calculation of the drift coefficient $\mathbf{A}$ and the diffusion coefficient D , the modified diffusion coefficient $\tilde{D}$ is calculated and the velocity update is performed the same way as in the diatomic case. The extension of the model lies in the rate coefficients as they indirectly model the internal energy structure of the molecules as well as the relaxation process. The coefficients need to be extended to account for an arbitrary number of degrees of freedom. The precise changes depending on the model of the rate coefficient which are discussed below.

Algorithm 1: Polyatomic

InOut : list of cells C
Input : species s , VHS parameters Ω , Δt

```
1 for each cell  $c$  in  $C$  do
2    $T \leftarrow \text{fn}(c.P)$ ; //  $c.P \rightarrow$  list of particles in cell  $c$ 
3    $v \leftarrow \text{fn}(c.P, \Omega)$ ; //  $v \rightarrow$  collision frequency
4    $K_{\text{rot}} \leftarrow \text{e\_transfer\_rot}(c.P, s, v, T, \Delta t)$ ;
5    $K_{\text{vib}} \leftarrow \text{e\_transfer\_vib}(c.P, s, v, T, \Delta t)$ ;
6    $A \leftarrow \text{fn}(c.P)$ ;
7    $D \leftarrow \text{fn}(c.P)$ ;
8    $\tilde{D} \leftarrow \sqrt{D^2 + 2(K_{\text{rot}} + K_{\text{vib}})}$ ;
9   for each particle  $p$  in  $c.P$  do
10     $p.v \leftarrow \text{fn}(p.v, A, \tilde{D})$ ;
11  end
12 end
```

Rotational Relaxation

For the relaxation of rotational energy a continuous energy structure is assumed and the extended CLT method employed. The difference between the diatomic and polyatomic case in the algorithm is the calculation of the parameter β (see Algorithm 2), which now includes d_{rot} as the number of rotational degrees of freedom to accommodate for non-linear molecules.

Algorithm 2: Rotational Relaxation - CLT

Input : list of particles P , species s , collision frequency v , T , Δt
Output: K

```
1  $e_{\text{rot}} \leftarrow 0$ ; // total rotational energy in cell
2  $\beta \leftarrow 2/(s.d_{\text{rot}} \cdot k_B \cdot T)$ ; //  $d_{\text{rot}} \rightarrow$  rotational degrees of freedom
3  $\tau_{\text{rot}} \leftarrow s.Z_{\text{rot}}/v$ ; //  $Z_{\text{rot}} \rightarrow$  rotational collision number
4  $P_{\text{rot}} \leftarrow 1 - \exp(-\Delta t/\tau_{\text{rot}})$ ;
5 for each particle  $p$  in  $P$  do
6   if random(0, 1) <  $P_{\text{rot}}$  then
7      $p.e_{\text{rot}} \leftarrow -\log(\text{random}(0, 1))/\beta$ ;
8   end
9    $e_{\text{rot}} \leftarrow e_{\text{rot}} + p.e_{\text{rot}}$ ;
10 end
11  $e_{\text{rot,ave}} \leftarrow e_{\text{rot}}/P.\text{size}$ ;
12  $e_{\text{rot}}^{\text{eq}} \leftarrow 1/\beta$ ;
13  $K \leftarrow -(e_{\text{rot}}^{\text{eq}} - e_{\text{rot,ave}})/(3 \cdot s.m \cdot \tau_{\text{rot}})$ ; //  $m \rightarrow$  species mass
```

Vibrational relaxation

Compared to the original implementation the difference in the new approach is that we now iterate over several vibration modes in contrast to only one that is present for diatomic molecules. Additionally, the degeneracy of vibrational modes must now be taken into account. At each step where the energy of a vibrational mode is used it needs to be multiplied by its degeneracy. For the *DLT* rate coefficient the modified approach is displayed in Algorithm 3.

For the *DLB* algorithm also the look-up-table holding the *DLB* parameters must be extended to account for different vibrational modes. This is achieved by increasing the dimensionality of $\psi(i, j, \frac{T}{\theta})$ accordingly. The approach is presented in detail in Algorithm 4.

Algorithm 3: Vibrational relaxation - DLT

Input : list of particles P , species s , collision frequency ν , T , Δt
Output: K

```

1  $e_{\text{vib}} \leftarrow 0$ ; // total vibrational energy in cell
2 for each particle  $p$  in  $P$  do
3   for each vibrational mode  $m$  of  $s$  do
4      $\tau_{\text{vib}} \leftarrow s.Z_{\text{vib}}[m]/\nu$ ; //  $Z_{\text{vib}}[m] \rightarrow$  vibrational collision number of mode  $m$ 
5      $P_{\text{vib}} \leftarrow 1 - \exp(-\Delta t/\tau_{\text{vib}})$ ;
6     if  $\text{random}(0, 1) < P_{\text{vib}}$  then
7        $i_{\text{vib,old}} \leftarrow p.i_{\text{vib}}[m]$ ;
8        $i_{\text{vib,new}} \leftarrow -\log(\text{random}(0, 1)) \cdot T/s.\theta[m]$ ; //  $\theta[m] \rightarrow$  vibrational temperature of mode  $m$ 
9        $p.e_{\text{vib}} \leftarrow p.e_{\text{vib}} + s.g[m] \cdot (i_{\text{vib,new}} - i_{\text{vib,old}}) \cdot k_B \cdot s.\theta[m]$ ; //  $g[m] \rightarrow$  degeneracy of vibrational mode  $m$ 
10    end
11  end
12   $e_{\text{vib}} \leftarrow e_{\text{vib}} + p.e_{\text{vib}}$ ;
13 end
14  $e_{\text{vib}}^{\text{eq}} \leftarrow 0$ ;
15  $\tau_{\text{vib,ave}} \leftarrow 0$ ;
16 for each vibrational mode  $m$  of  $s$  do
17    $e_{\text{vib}}^{\text{eq}} \leftarrow e_{\text{vib}}^{\text{eq}} + s.g[m] \cdot k_B \cdot s.\theta[m] / (\exp[s.\theta[m]/T] - 1)$ ;
18    $\tau_{\text{vib,ave}} \leftarrow \tau_{\text{vib,ave}} + s.Z_{\text{vib}}[m]$ ;
19 end
20  $\tau_{\text{vib,ave}} \leftarrow \tau_{\text{vib,ave}}/s.v_N$ ; //  $v_N \rightarrow$  number of vibrational modes
21  $e_{\text{vib,ave}} \leftarrow e_{\text{vib}}/P.\text{size}$ ;
22  $K \leftarrow -(e_{\text{vib}}^{\text{eq}} - e_{\text{vib,ave}})/(3 \cdot s.m \cdot \tau_{\text{vib,ave}})$ ; //  $m \rightarrow$  species mass

```

TABLE 1. Inflow conditions for hypersonic flow around cylinder.

Velocity	Temperature	Number density	Mach number	Knudsen number
1862.6 ms^{-1}	90.1 K	$6 \times 10^{20} \text{ m}^{-3}$	12.77	0.0147

NUMERICAL RESULTS AND DISCUSSION

To validate the extension a code-to-code comparison was performed between the augmented Fokker-Planck implementation and a DSMC implementation done in the *PicLas* code at the University of Stuttgart [17]. The simulations use *CLT* rate coefficients for the rotational energy modes and the *DLT* model for the vibrational ones.

Heat Bath

The first test case is a heat bath in which a gas is initialized with different translational, rotational and vibrational temperatures ($T_{\text{trans}} = 1 \times 10^4 \text{ K}$, $T_{\text{rot}} = 7.5 \times 10^3 \text{ K}$, $T_{\text{vib}} = 5 \times 10^3 \text{ K}$). Subsequently the system is let to relax to equilibrium. The simulation was performed for two different gas species, namely CO_2 and CH_4 . The codes agree very well in both cases, as can be seen in Fig. 2 (CO_2) and Fig. 3 (CH_4).

Hypersonic, Non-Reactive Flow Over a Flat-Faced Cylinder

For a more sophisticated test case a non-reacting, hypersonic, CO_2 flow around a flat-faced cylinder is simulated. The setup can be seen in Fig. 4 where $R = 50 \text{ mm}$, $r = 12.5 \text{ mm}$. The computational domain has dimension of $50 \text{ mm} \times 75 \text{ mm}$. The simulation uses the axial symmetry of the problem with respect to the x -axis. Inflow conditions are reproduced in Table 1 [17].

Algorithm 4: Vibrational relaxation - DLB

Input : list of particles P , species s , collision frequency ν , T , Δt , DLB parameters D , VHS parameters Ω
Output: K

```

1  $e_{\text{vib}} \leftarrow 0$ ; // total vibrational energy in cell
2 for each particle  $p$  in  $P$  do
3   for each vibrational mode  $m$  of  $s$  do
4      $P_{\text{vib}} \leftarrow \text{fn}(s.Z_{\text{vib}}[m], \Omega, \omega, s.\theta[m], T)$ ; // calculated as for the diatomic case
5      $i_{\text{temp}} \leftarrow \text{fn}(D, T, s.\theta[m])$ ; // temperature id, calculated as for the diatomic case
6      $i \leftarrow p.i_{\text{vib}}[m]$ ; //  $i_{\text{vib}}[m] \leftarrow$  old energy level of mode  $m$ 
7      $j \leftarrow i$ ;
8      $t \leftarrow 0$ ;
9     while true do
10       $q \leftarrow D.\text{rates\_sum}[m][i][i_{\text{temp}}] \cdot \nu \cdot P_{\text{vib}}$ ; //  $\text{rates\_sum} \leftarrow \sum_j^N R_{ij,m}(i_{\text{temp}})$ 
11       $R \leftarrow \text{random}(0, 1)$ ;
12       $t \leftarrow t - (\log(R)/q)$ ;
13      if  $t > \Delta t$  then
14        break;
15      end
16      while do
17         $j \leftarrow \text{random}(0, 1) \cdot D.N_s$ ; //  $N_s \rightarrow$  total number of states
18        while  $j = i$  do
19           $j \leftarrow \text{random}(0, 1) \cdot D.N_s$ ;
20        end
21        if  $D.\text{rates}[m][i][j][i_{\text{temp}}] > \text{random}(0, 1)$  then
22          //  $\text{rates} \leftarrow$  look up table for rate constants  $R_{ij,m}(i_{\text{temp}})$ 
23          break;
24        end
25         $p.i_{\text{vib}}[m] \leftarrow j$ ;
26      end
27       $p.e_{\text{vib}} \leftarrow p.e_{\text{vib}} + s.g[m] \cdot (j - i) \cdot k_B \cdot s.\theta[m]$ ; //  $g[m] \rightarrow$  degeneracy of vibrational mode  $m$ 
28    end
29     $e_{\text{vib}} \leftarrow e_{\text{vib}} + p.e_{\text{vib}}$ ;
30 end
31  $e_{\text{vib}}^{\text{eq}} \leftarrow 0$ ;
32  $\tau_{\text{vib,ave}} \leftarrow 0$ ;
33 for each vibrational mode  $m$  of  $s$  do
34    $e_{\text{vib}}^{\text{eq}} \leftarrow e_{\text{vib}}^{\text{eq}} + s.g[m] \cdot k_B \cdot s.\theta[m] / (\exp[s.\theta[m]/T] - 1)$ ;
35    $\tau_{\text{vib,ave}} \leftarrow \tau_{\text{vib,ave}} + s.Z_{\text{vib}}[m]$ ;
36 end
37  $\tau_{\text{vib,ave}} \leftarrow \tau_{\text{vib,ave}} / s.v_N$ ; //  $v_N \rightarrow$  number of vibrational modes
38  $e_{\text{vib,ave}} \leftarrow e_{\text{vib}} / P.\text{size}$ ;
39  $K \leftarrow -(e_{\text{vib}}^{\text{eq}} - e_{\text{vib,ave}}) / (3 \cdot s.m \cdot \tau_{\text{vib,ave}})$ ; //  $m \rightarrow$  species mass

```

Displayed in Fig. 5 and Fig. 6 are the temperature distributions produced by the Fokker-Planck code and the reference DSMC implementation, respectively. The color sets are chosen to be equal in both pictures. At first glance both results are reproducing the shock shape very well. The same can be said for the number density and to some extent for the velocity in x -direction. Both are plotted along the axis of symmetry in Fig. 7 and Fig. 8 respectively.

Closer inspection of the velocity profile and especially the temperature distribution over the symmetry axis (Fig. 9) however reveals a noticeable deviation around the position of the shock, especially at its front. This behaviour has already been observed [11] and the reason for the deviation was attributed to the constant time scale τ used in the Fokker-Planck formulation. One also need to consider the general validity of the Fokker-Planck formulation in the area in and close to the shock as it is far away from translational thermal equilibrium, which is an assumption made in the derivation of the drift and diffusion coefficients. In practice for our application case of a hybrid code this is not considered to be an issue. The final goal of the Fokker-Planck development at DLR Goettingen is to create a method which will use DMSC in areas of strong thermal non equilibrium and Fokker-Planck in areas close to the continuum

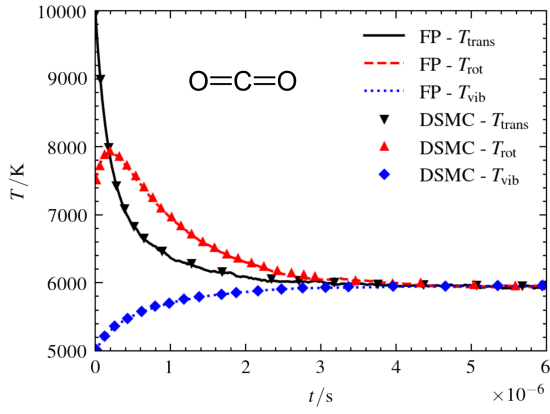


FIGURE 2. CO2 heat bath.

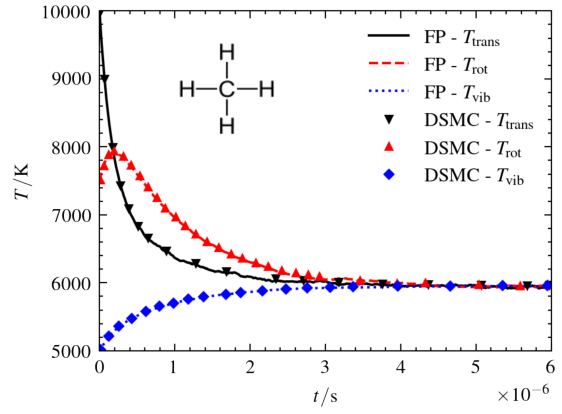


FIGURE 3. CH4 heat bath.

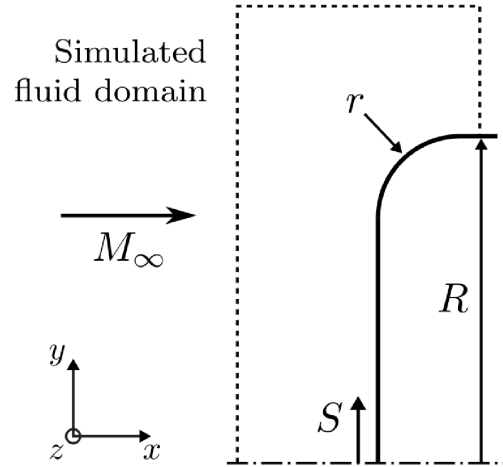


FIGURE 4. Simulation Domain [17].

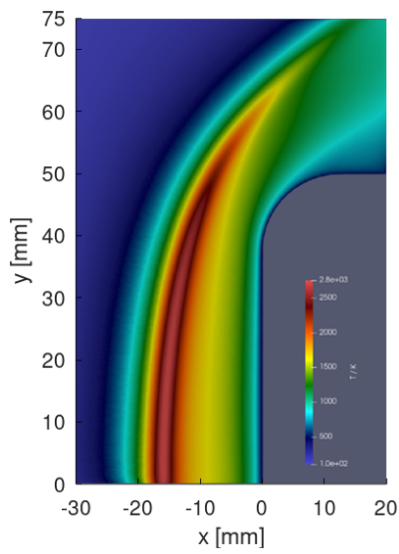


FIGURE 5. Fokker-Planck temperature.

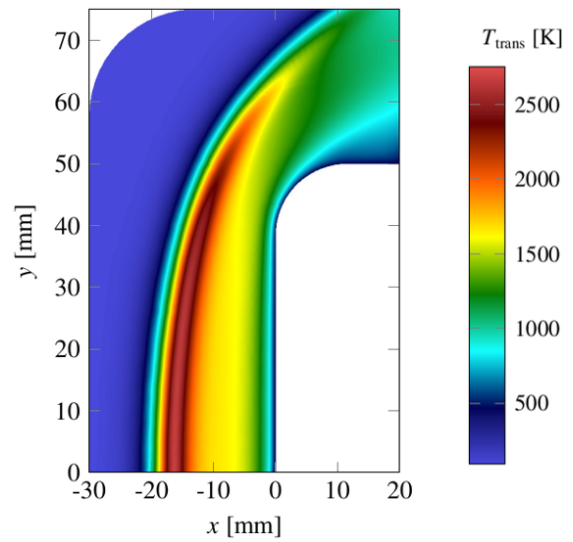


FIGURE 6. DSMC temperature.

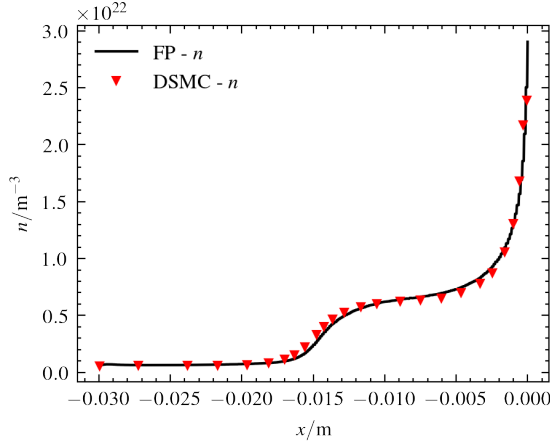


FIGURE 7. Number density.

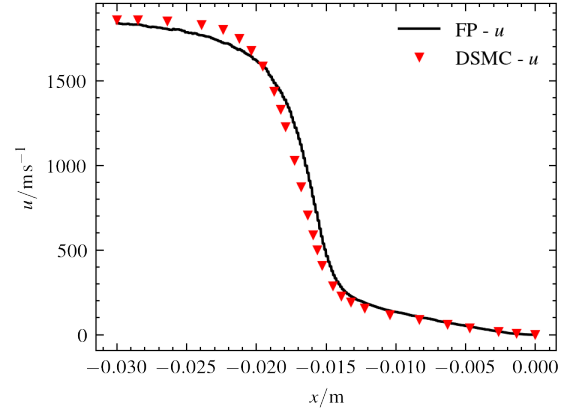


FIGURE 8. Velocity.

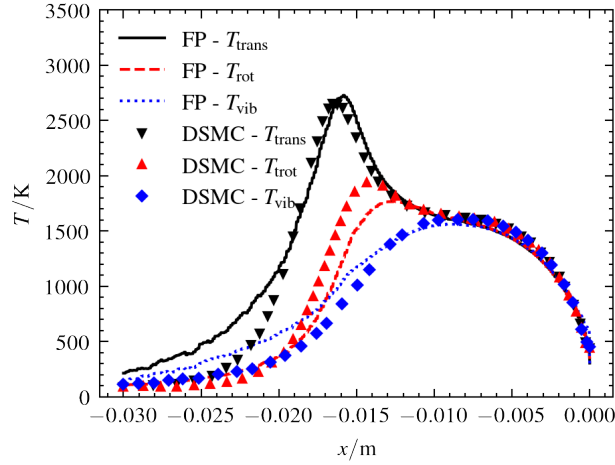


FIGURE 9. Temperature.

regime.

DISCUSSION AND CONCLUSION

An extension of the Fokker-Planck method for polyatomic gases using the master equation approach is presented. The model is compatible with different Fokker-Planck formulations. An example using the cubic formulation is presented. Code-to-code comparison between the new model and the DSMC implementation in PicLas code of the university of Stuttgart is presented using a heat bath temperature relaxation as well as two dimensional hypersonic flow around a flat-faced cylinder. The heat bath test case provides very good agreement between the DSMC and Fokker-Planck implementation. The hypersonic test compares very well qualitatively but as expected shows deviations in areas of strong thermal non-equilibrium.

The derivation of the polytomic model is based on the assumption that no energy transfer between vibration and rotation (V-R) and between the individual vibration modes takes place (V-V). This assumption is motivated by experimental data for CO₂ which shows that exchanges within the same vibrational mode occur much more often than V-V and V-R exchanges [18]. Therefore the assumption is made that V-V and V-R can be modeled via the indirect energy transfer through the translational energy as it happens on a slower time scale.

However, this assumption can not be made for all cases. Using N₂ as an example it can be expected that for

high temperatures vibrational-rotational transitions will occur more frequently through mechanism like resonance exchange and simultaneous rotational and vibrational excitation/de-excitation [19]. According to Park [19] such fast vibrational-rotational exchange will occur when the high vibrational and rotational levels are populated significantly. Additionally it should be noted that physical effects like unharmonic pumping can not be modeled due the assumption of equal energy spacing based on the harmonic oscillator. This makes the presented approach inapplicable to the simulation of certain devices, for example gas dynamic lasers [20].

In previous work by Gorji et. al. the Fokker-Planck approach was extended to the so called entropic model which correctly reproduces entropy production rates. It was shown that the new model better predicts flows in areas of high thermal non-equilibrium compared to cubic Fokker-Planck [21]. As the master equation approach separates the treatment of the internal degrees of freedom from the translational velocities it could potentially be extended to be used with the entropic formulation of Fokker-Planck. The so extended model could reduce the differences present in this work between the Fokker-Planck and DSMC results. This will be investigated in future work.

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